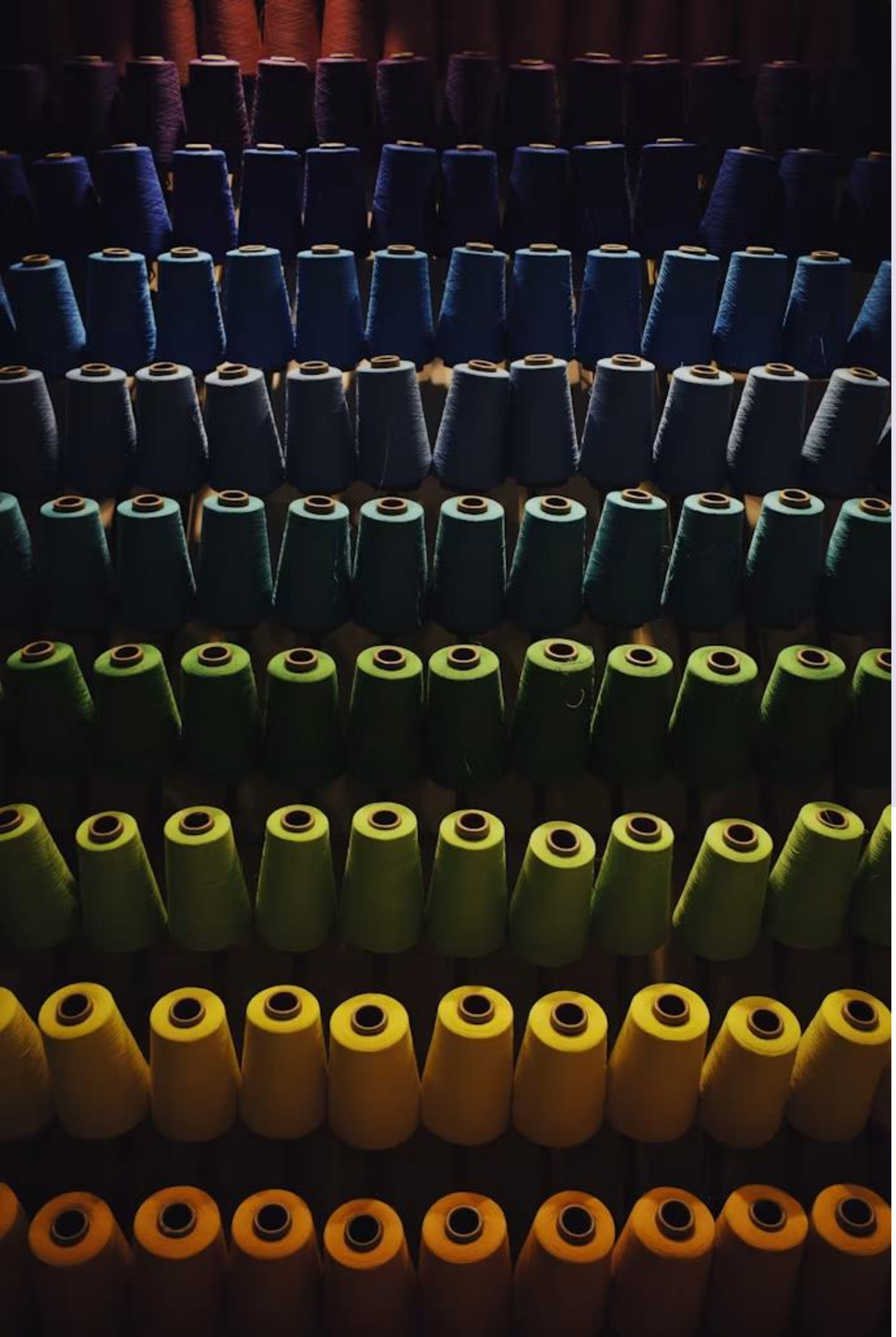


FABRIC MANUFACTURING

Part: A

Course Code: FE 351-0723

Mechanical Principles of Knitting Technology

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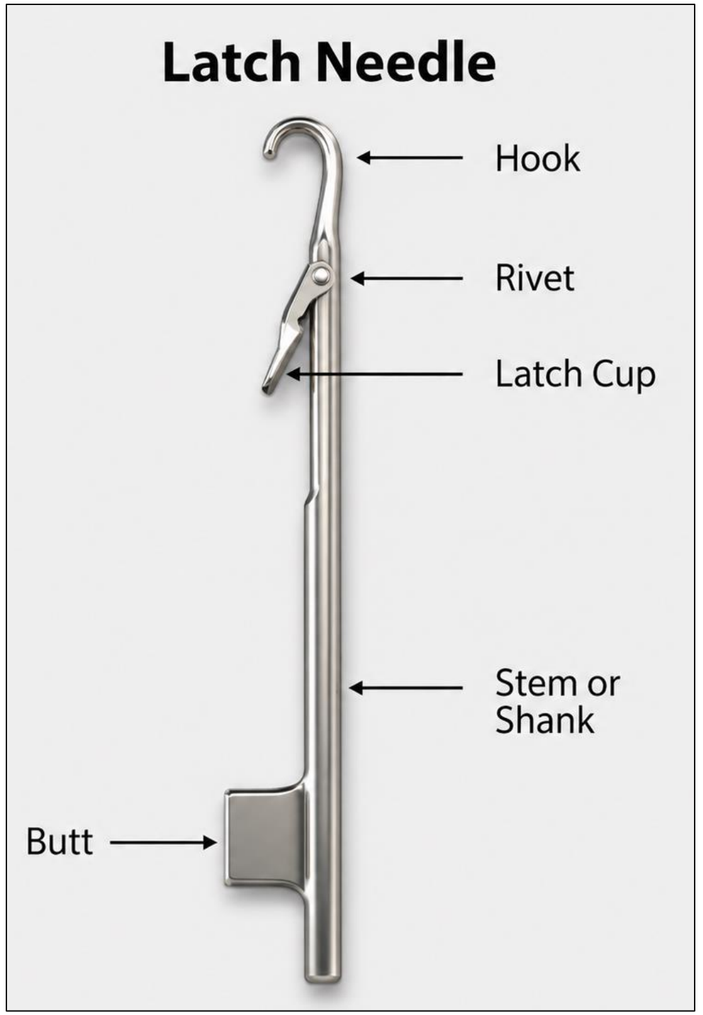
E-mail: arnob@fe.butex.edu.bd

BASIC ELEMENTS

Needle

The primary knitting element in knitting machine is needle which makes a compound movement for producing loops. In circular weft knitting, the needle moves round the machine axis along with the cylinder and at the same time performs an upward and downward movement. But in flat-bed knitting machine, needle is fixed onto the needle bed. Needles are mainly of three types.

1. Latch needle
2. Spring bearded needle
3. Compound needle.



DIFFERENT TYPES OF NEEDLE



Latch Needle for CKM



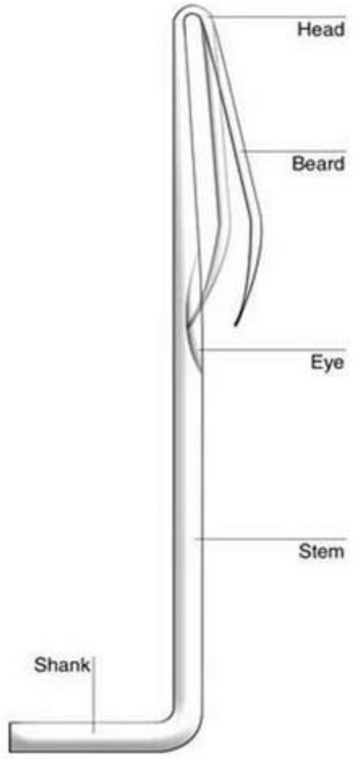
Latch needle for FKM



DE Latch Needle for PKM



Latch Needle for Warp KM



Bearded Needle for Warp KM



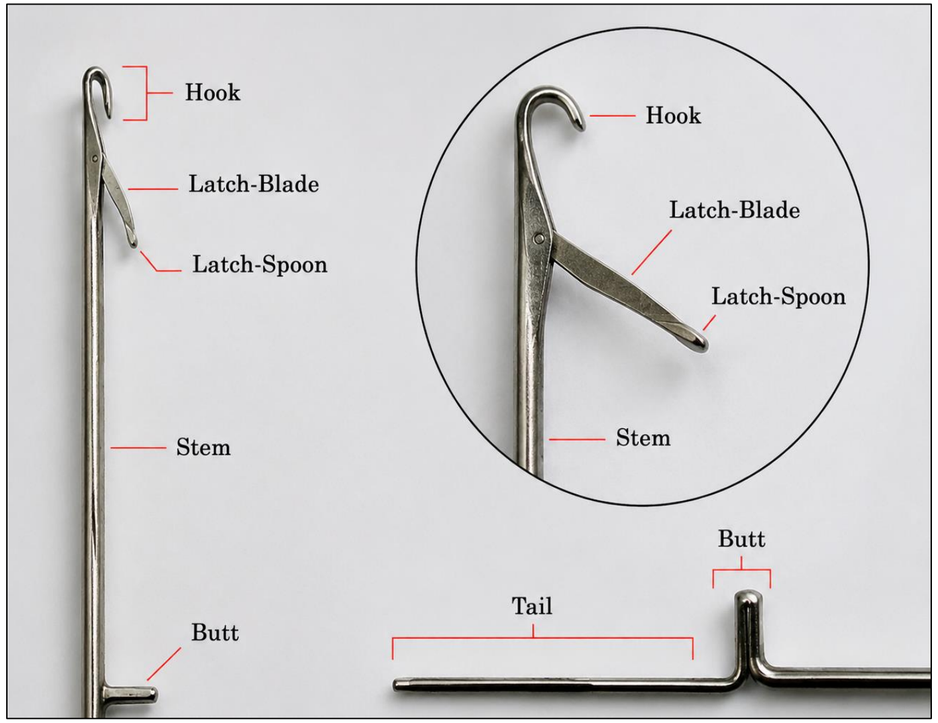
Compound Needle for Warp KM

NEEDLES

Latch Needle

The latch needles have following parts:

1. Hook: Grasping a new yarn in the knitting process.
2. Latch: Freely rotating around the axle.
3. Latch Blade: Locates the latch in the needle.
4. Latch Spoon: Extension of latch blade bridges the gap between hook when closed.
5. Stem: Carries the loop in clearing position.
6. Rivet: This has been dispensed with on most plate metal needles by pinching in the slot walls to retain the latch blade.
7. Butt: Which serving to displace the needle along the needle bed slot.
8. Tail: Extension below the butt giving additional support.



NEEDLES

Spring Bearded Needle

The spring-bearded needles made of steel wire consist of the following parts:

a. The Stem: The stem around which the needle loop is formed.
b. The Head: Where the stem is turned into a hook to draw the new loop through the old loop.
c. The Beard: Which is the curved downwards continuation of the hook that is used to separate the trapped new loop inside from the old loop as it slides off the needle beard.
d. The Eye or groove: Cut in the stem to receive the pointed tip of the beard when it is pressed, thus enclosing the new loop.
e. Shank: Serving to attach the needle in the needle bar.

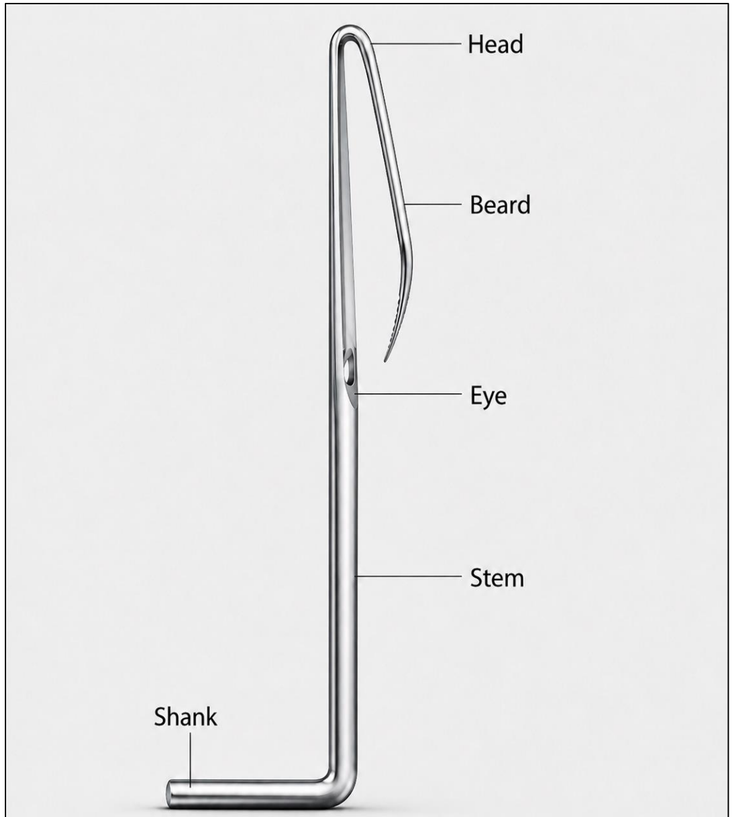


Fig: Spring Bearded needle

NEEDLES

Compound Needle

- Compound Needle consists of two separately controlled parts; these are- the open hook and the hook closing element (tongue, latch, piston, and plunger).
- The two parts rise and fall as a single unit but at the top of the rise, the hook moves faster to open and at the start of the fall the hook descends faster to close the hook.
- Knitting action is short, smooth, and simple. So, the production rate is the highest for this needle. Also, yarn is subjected to lowest strain.
- It is easier to drive the hooks and tongues collectively from two separate bars as in warp knitting since no presser or rivet are used like bearded and latch needle respectively. It is widely used in warp knitting due to the leverage of attaining higher knitting speed.
- Manufacturing cost is higher.



BASIC ELEMENTS

Cam:

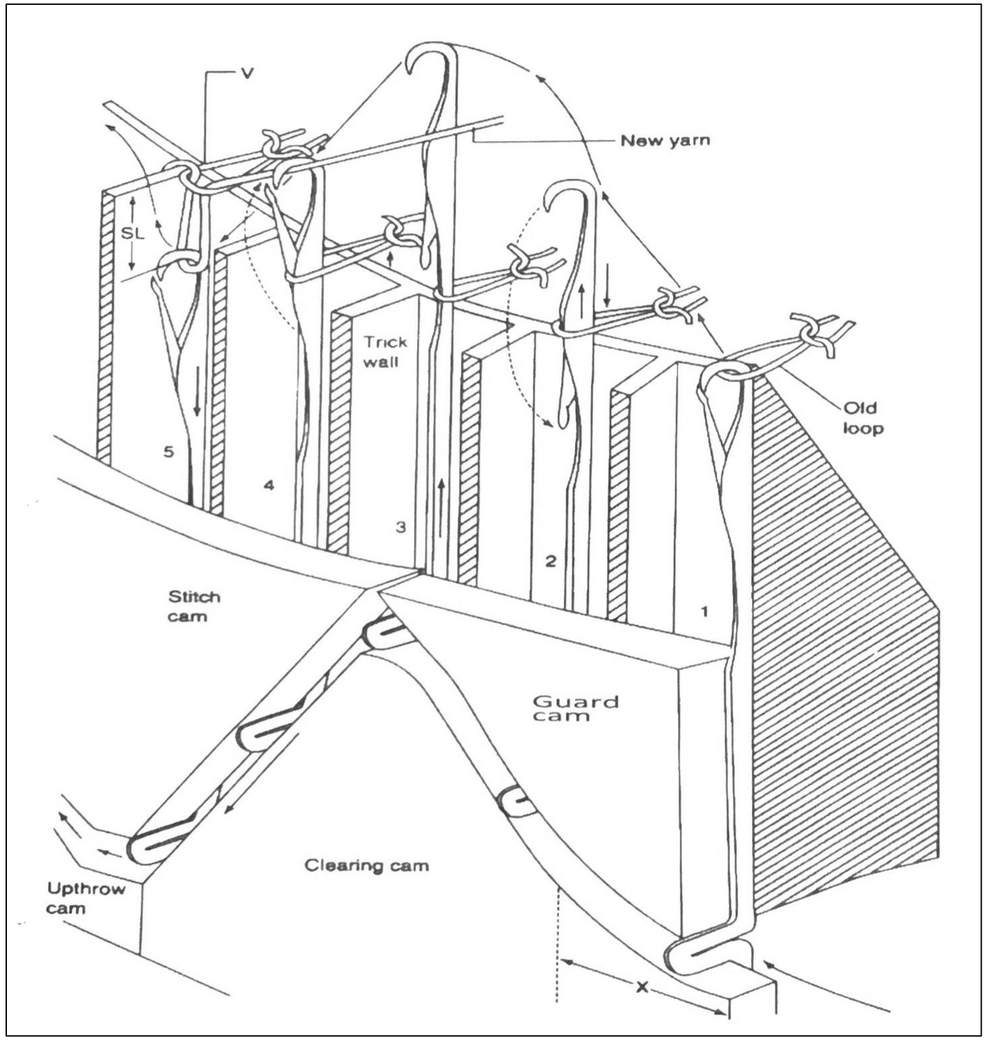
Cam is the third primary knitting elements which convert the rotary machine drive into a suitable reciprocating action to the needles or other elements. Cam assists the needle in loop formation. Knitting cams are of three types:

1. Knit cam
2. Tuck cam
3. Miss cam



Knit Cam | Tuck Cam | Miss Cam

PARTS OF A KNIT CAM



1. <u>The Clearing Cam:</u> This causes the *needles to be lifted* to either *tuck*, *clearing loop transfer* or needle transfer height depending upon machine design. *Clearing/Raising cam is mainly classified into knit, tuck, and miss.*

2. <u>The Stitch Cam:</u> It controls *the depth to which the needle descends* thus controlling the *amount of yarn drawn into the needle loop.*

3. <u>The Up-throw Cam:</u> It takes the *needles back to the rest position* and allows the newly formed loops to relax.

4. <u>The Guard Cam:</u> These are often placed on the opposite side of the cam race to *limit the movement of the butts and to prevent needles from falling out of track*.

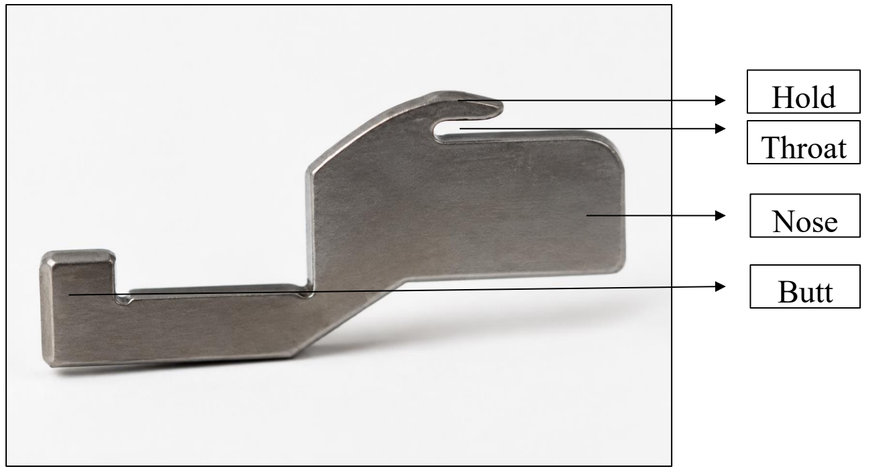
BASIC ELEMENTS

Sinker

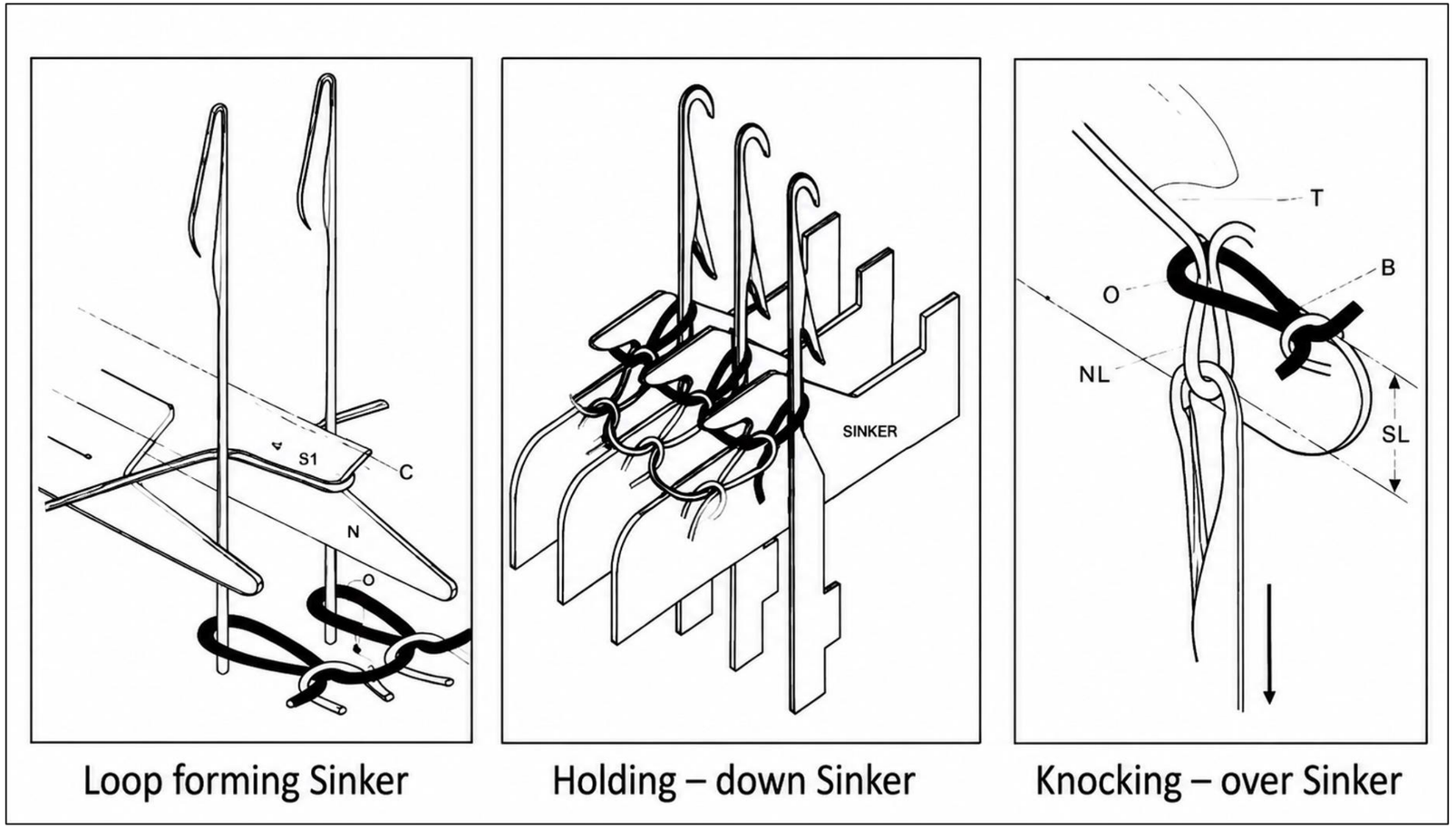
Sinker is the second primary knitting element though double jersey knitting machines do not use sinker. It is a specially shaped flat metal plate that sits in between each needle in single jersey knitting machine and warp knitting machine. The sinker helps in loop formation, holding down and knocking over (the needle passes through the old loop by completely drawing a new loop). Sinkers are placed horizontally.

It is mainly of three types:

1. Loop Forming sinker (Bearded needle-based weft knitting machine straight bar frame)
2. Holding down sinker (CKM)
3. Knocking over (Warp knitting)



BASIC ELEMENTS



BASIC ELEMENTS

1. Loop Formation
   - ❖ The first function of a sinker is to *help in forming a new loop.*
   - ❖ In bearded needle weft knitting machines, *the sinker pushes or bends the newly supplied yarn downward between two adjacent needles to form a loop.*
   - ❖ However, *in latch needle weft knitting* and warp knitting machines, *sinkers do not take part in loop formation.*

BASIC ELEMENTS

2. Holding Down Old Loops
   - ❖ The most important function of a modern sinker is to hold down the old loops during knitting.
   - ❖ When needles move upward to form new loops, the *sinker prevents the old loops from rising with the needles.*
   - ❖ The nose or nib of the sinker holds the old loops in their position and keeps the fabric stable.

This function helps to produce:

Better fabric appearance, Tighter knitted structures, Higher knitting speed, Lower fabric draw-off tension.

In Tricot warp knitting and single bed weft knitting machines, a slot or *throat in the sinker helps to control the old loop.* In Raschel warp knitting machines, the *sinker works mainly as a web holder or stitch comb.*

BASIC ELEMENTS

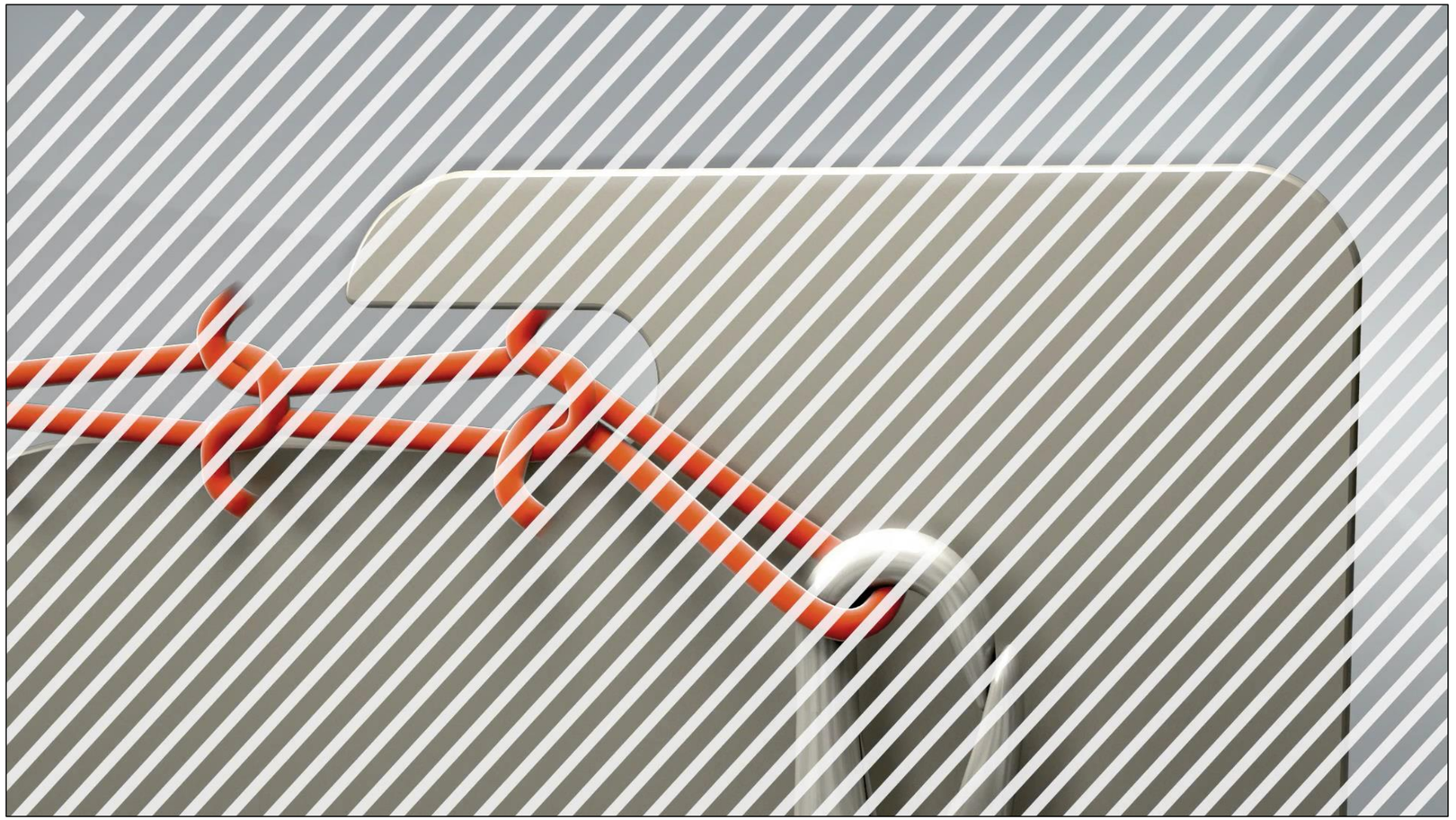
3. Knock-over Function

- The sinker also works as a knock-over surface during stitch formation.
- The upper surface (called the belly) *supports the old loop while the new loop is pulled through it.*
- In tricot warp knitting machines, *the sinker belly is specially designed to help in landing and knock-over.*
- In latch needle machines, the *upper surface of the trick plate* performs the *knock-over function*.

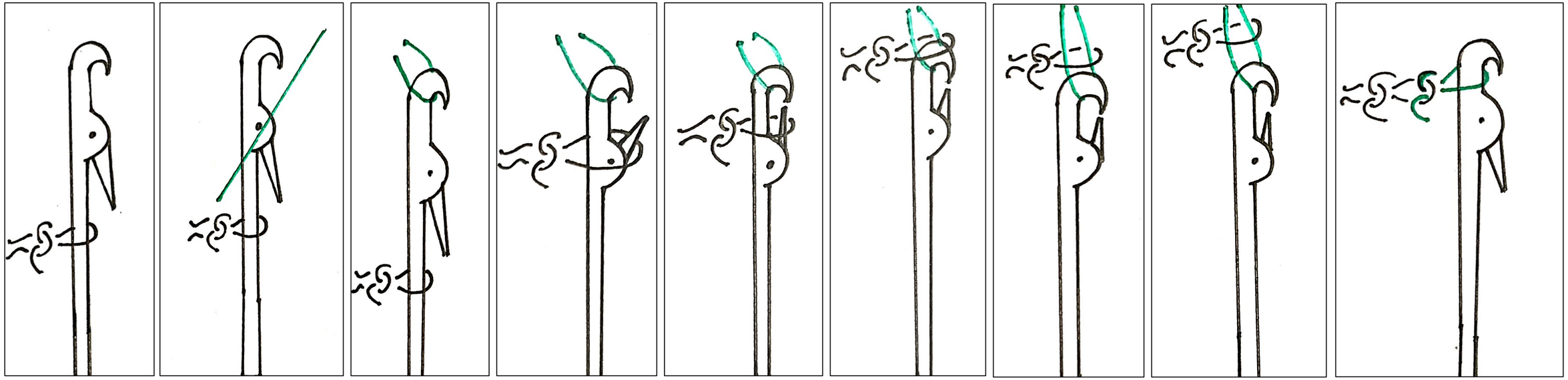
KNITTING CYCLE OF LATCH NEEDLE

1. Clearing: The process of stitch formation is started by the clearing operation. Its aim is to *draw the old loops behind the needle latch into the needle stem*. Clearing is completed when the *needle reaches its upper position.*
2. Yarn feeding: In the course of needle downward motion, the hook takes hold of yarn coming from the yarn guide.
3. Underlapping: Further movement of *yarn just laid* and its *forwarding under the needle hook* is called under lapping.
4. Pressing: The aim of *pressing is to close the needle hook* with the yarn laid in it.
5. Landing: The essence of this operation consists in *shifting the old loop on the closed latch.*
6. Joining: At joining, the *new yarn comes in contact with the old loop* but remain in the exposure of the hook of the needle.
7. Knock over: *Old loop completely leaves the needle* and intermeshes with the new loop.
8. Stich formation: The needle further descends *depending on the depth of the stich cam* and a fully *completed loop is formed.*
9. Loop Draw-off: The aim of this operation is to *draw the old loop behind the needle back.* It is *affected by sinker throats.*

KNITTING CYCLE OF LATCH NEEDLE



KNITTING CYCLE OF LATCH NEEDLE



CLEARING

YARN RECEIVING

UNDER LAPPING

PRESSING

LANDING

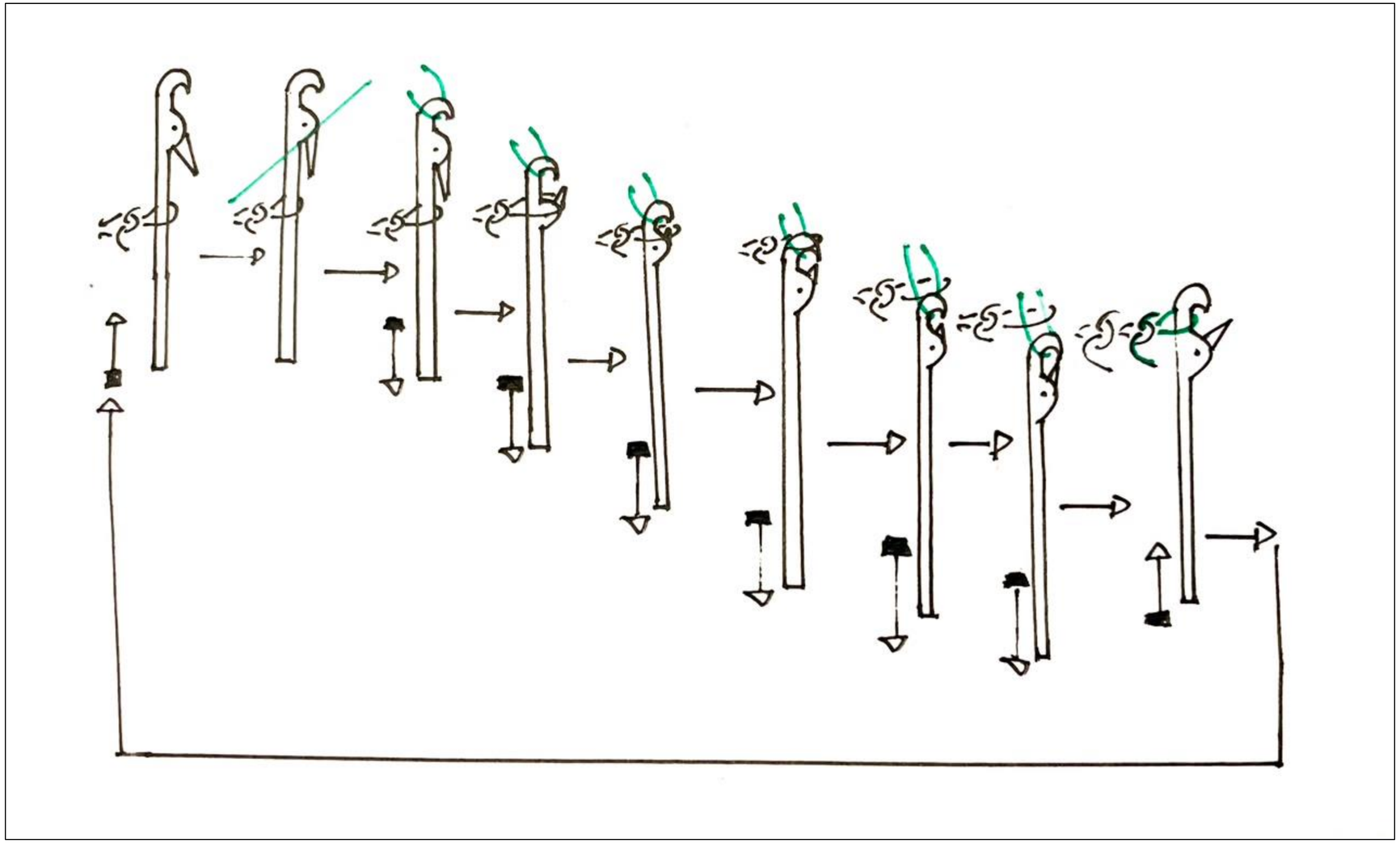
JOINING

KNOCKING OVER/CASTING OFF

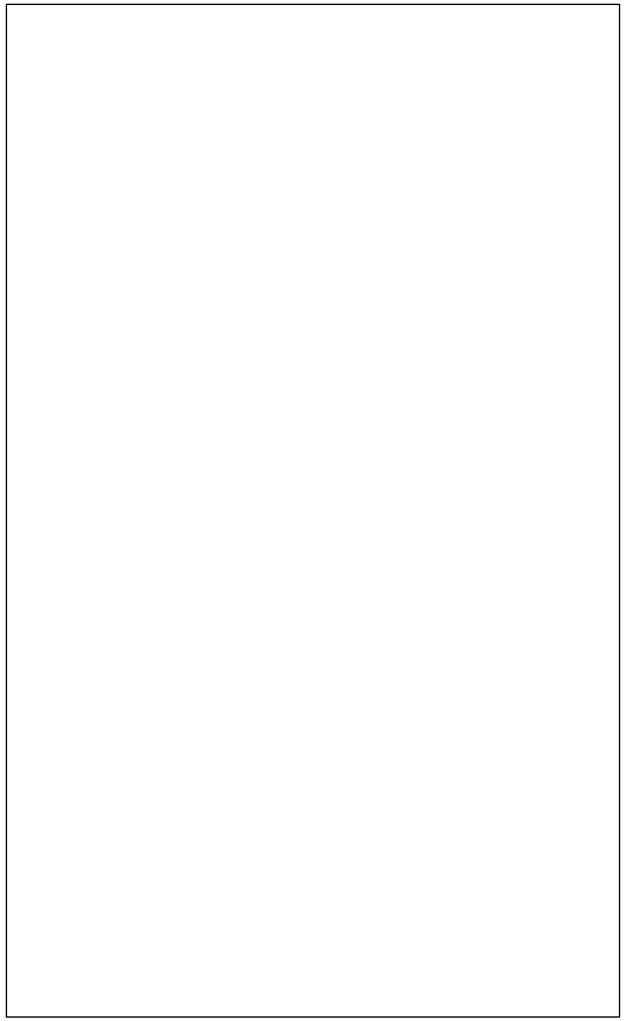
LOOP FORMING

LOOP DRAW OFF

KNITTING CYCLE OF LATCH NEEDLE



KNITTING CYCLE OF LATCH NEEDLE



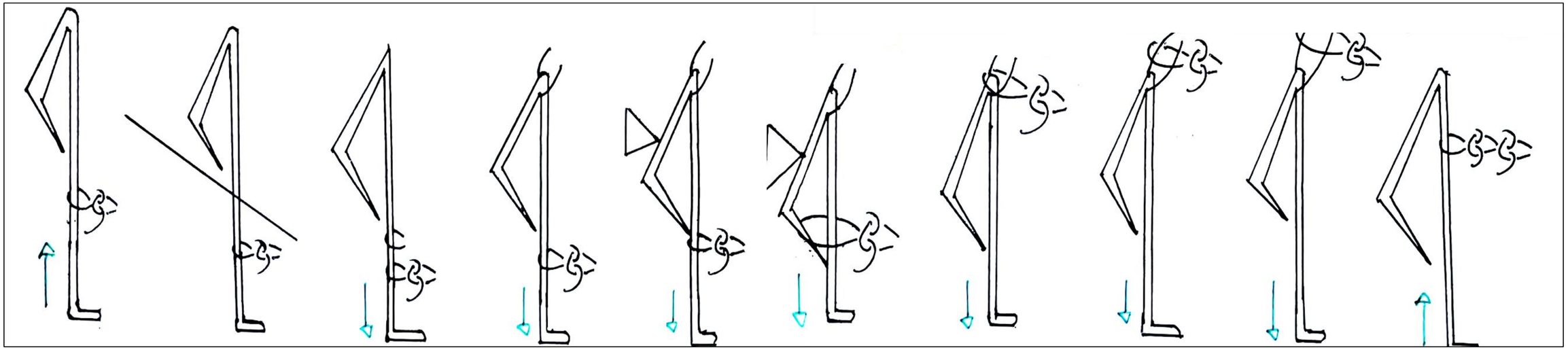
KNITTING CYCLE OF SPRING BEARDED NEEDLE

1. Clearing: Needle is placed *over the top-most position.* The hook is open, and the *old loop is placed along the needle stem* and again started to form new loop.
2. Yarn feeding: New yarn is feed under the *throat of kinking sinker.*
3. Yarn Sinking: The new *yarn is transformed into loop shape by using throats of loop forming sinker*. It is the disadvantage of bearded needle.
4. Underlapping: New loop is formed in the hook.
5. Pressing: Needle *beard is pressed in* the needle grove by *pressure disc.*
6. Landing: The *old loop lands on the beard*. Simultaneously *needle continues to move downwards.*
7. Joining: The *old loop gets closer the new loop* as the needle descends but remain in the exposure of the hook of the needle.
8. Casting-off Knocking over: Old loop leaves the needle and intermeshes with the new loop.
9. Stich formation: The needle further descends depending on the depth of the stich cam and a fully completed loop is formed.
10. Loop Draw-off: *Due to tension of take down roller, loops become horizontal from vertical position.*

KNITTING CYCLE OF SPRING BEARDED NEEDLE



KNITTING CYCLE OF SPRING BEARDED NEEDLE



CLEARING | YARN RECEIVING | YARN SINKING | UNDER LAPPING | PRESSING | LANDING | JOINING | KNOCKING OVER/CASTING OVER | LOOP FORMING | LOOP DRAW OFF

BASIC KNITTING STEPS

Basic knitting steps of Bearded needle: (4 Steps)

1. Yarn Feeding or Laying
2. Yarn Sinking or Loop Formation
3. Beard Pressing or Hook Closing
4. Knocking over or Casting off

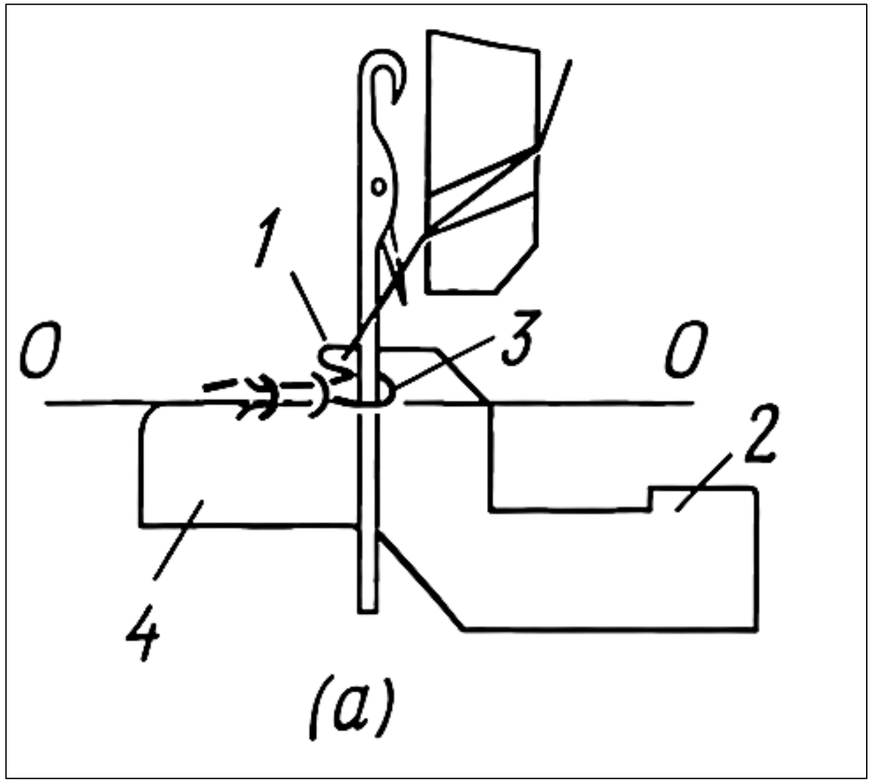
Basic knitting steps of Latch needle: (3 Steps)

1. Yarn Feeding or Laying
2. Pressing or Hook Closing
3. Knocking over or Casting off

KNITTING CYCLE OF LATCH NEEDLE

Clearing (a)

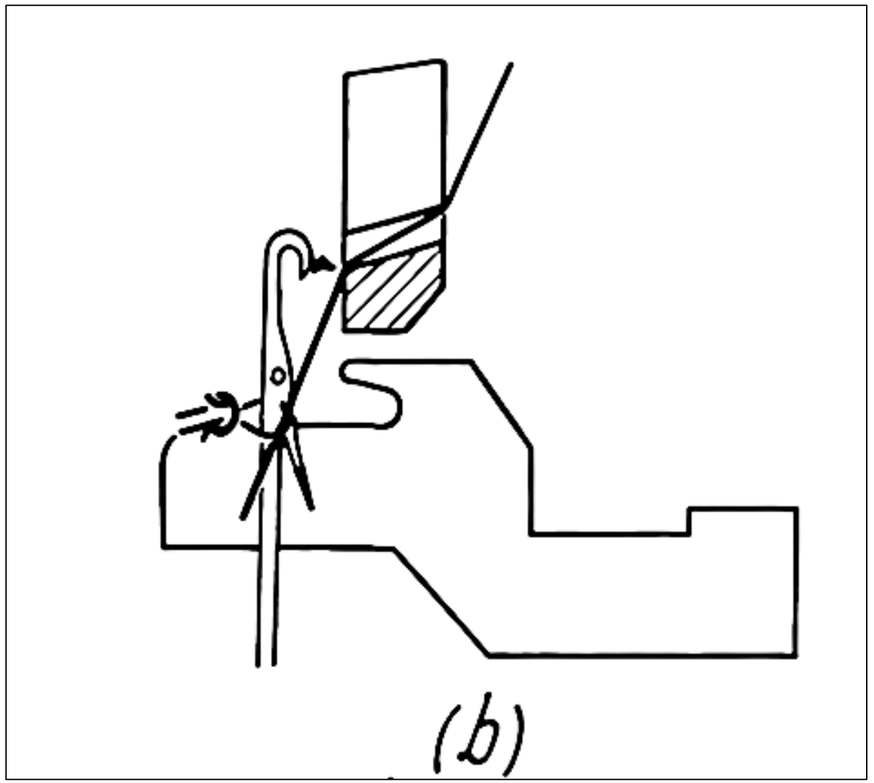
- Clearing is the *first step* of making a stitch.
- In this step, the needle moves upward.
- The *old loop moves behind the needle latch.*
- The *sinker holds the old loop so that it does not move upward with the needle.*
- During clearing, *the sinker stays still.*
- *The clearing process finishes when the needle reaches its highest position.*



KNITTING CYCLE OF LATCH NEEDLE

Yarn Feeding (b)

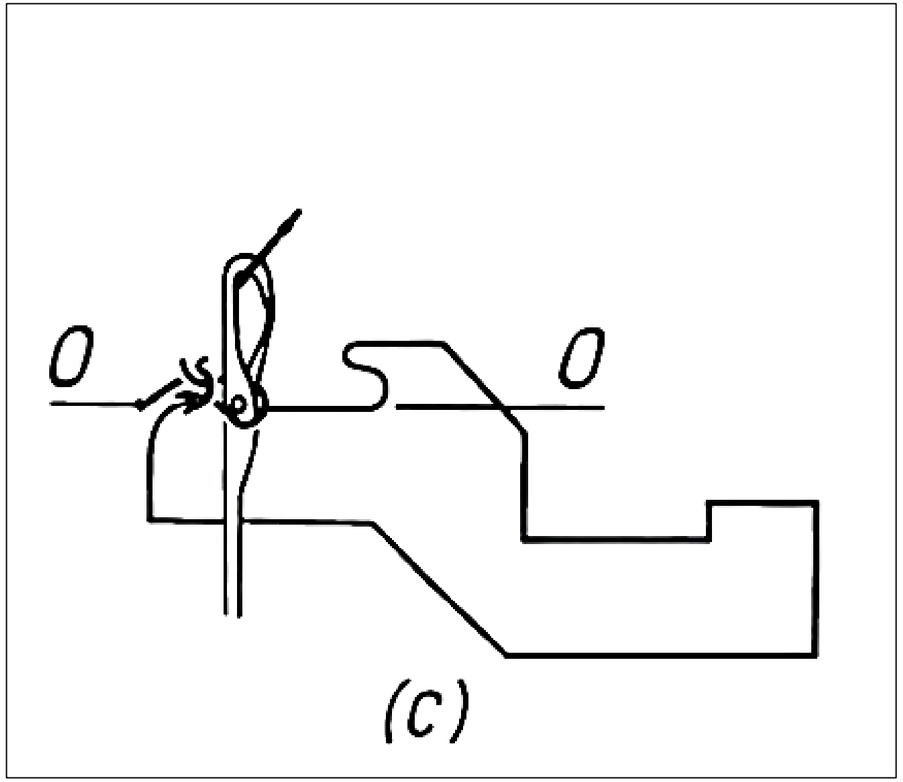
- Yarn laying is the *second step* of stitch formation.
- In this step, the *needle moves downward.*
- The *hook catches the yarn coming from the yarn guide.*
- The hook places the yarn around the needle to form a new loop.
- When yarn laying starts, the *sinker moves slightly away from the cylinder center.*
- This movement *gives enough space for the loop to form easily.*



KNITTING CYCLE OF LATCH NEEDLE

Underlapping (c)

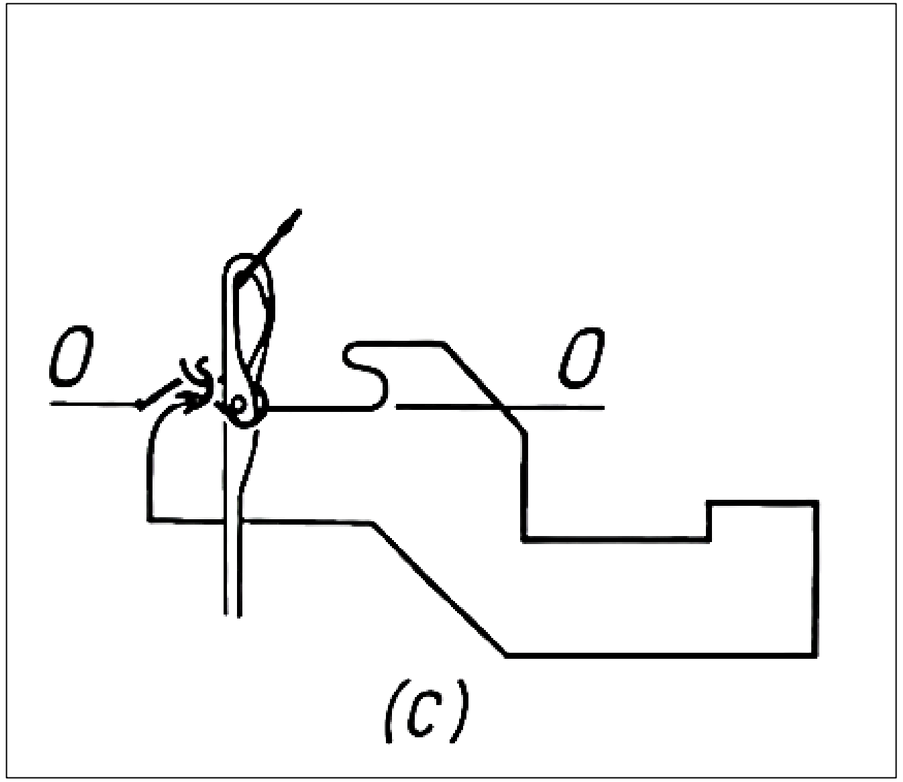
- Under lapping is the *third step* of stitch formation.
- After the yarn is placed around the needle, *it is moved forward under the needle hook.*
- This *movement of the yarn is called under lapping or yarn drawing.*
- It *helps to position the yarn* correctly for *making the new stitch.*
- Under lapping happens at the *same time as yarn laying when the needle starts moving downward.*



KNITTING CYCLE OF LATCH NEEDLE

Pressing (c)

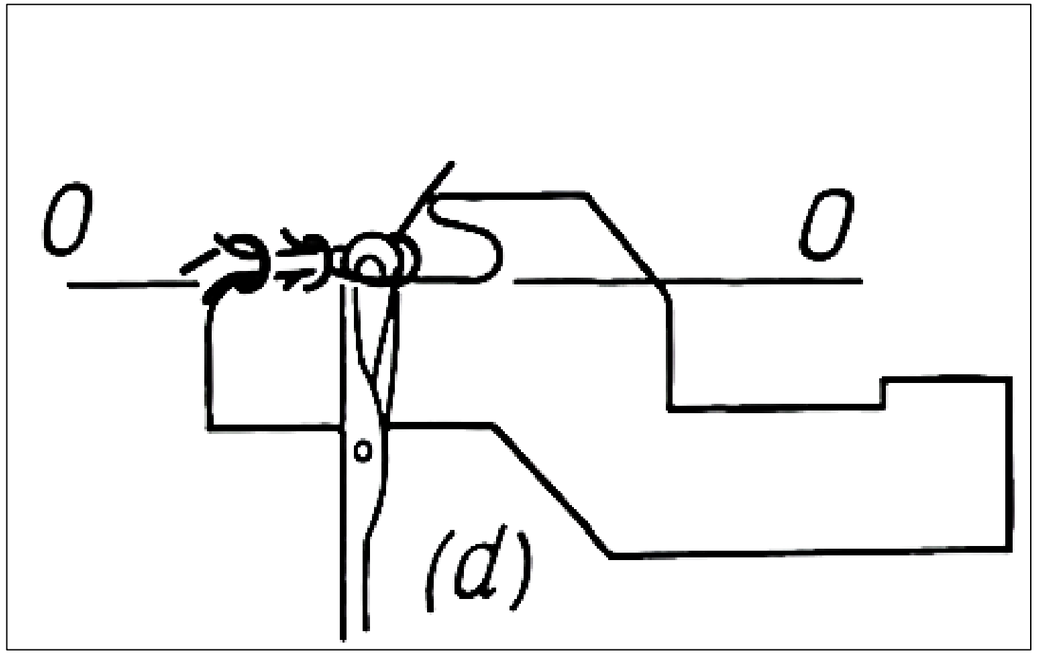
- Pressing is the *fourth step* of stitch formation.
- The main purpose of pressing is *to close the needle hook with the yarn inside it.*
- When the needle moves downward, the *latch touches the old loop.*
- The old *loop is held by the sinker chin* and cannot move down.
- The old loop pushes the latch and closes the needle hook.



KNITTING CYCLE OF LATCH NEEDLE

Landing (d)

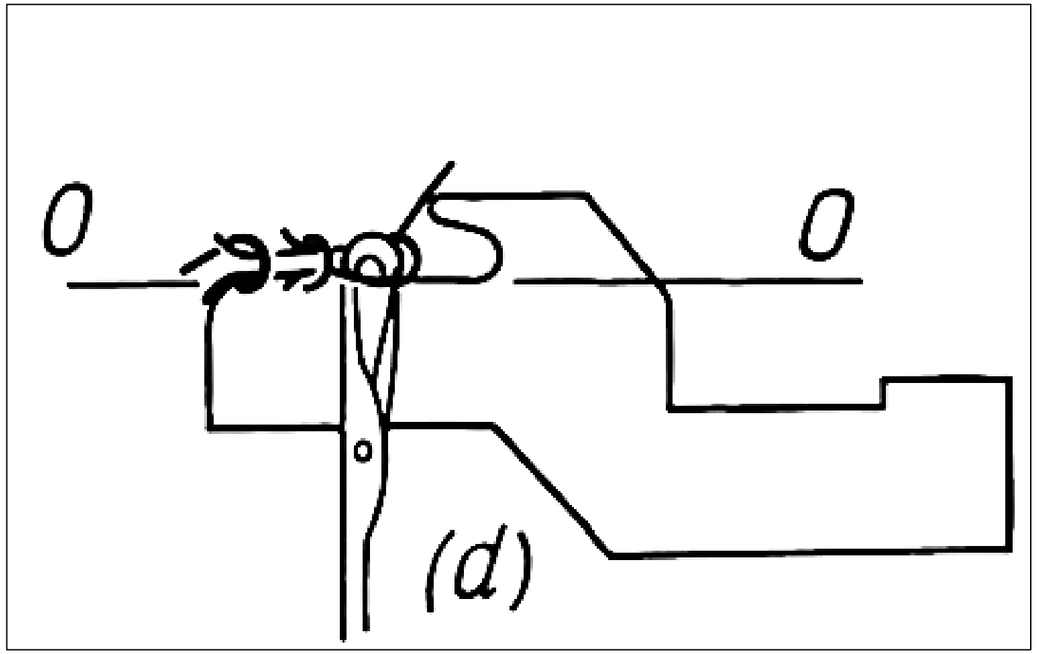
- Landing is the next step after pressing.
- In this step, the *old loop moves onto the closed latch of the needle.*
- The *old loop slides over the closed latch.*
- In latch needle machines, *landing starts at the same time as pressing.*



KNITTING CYCLE OF LATCH NEEDLE

Joining (d)

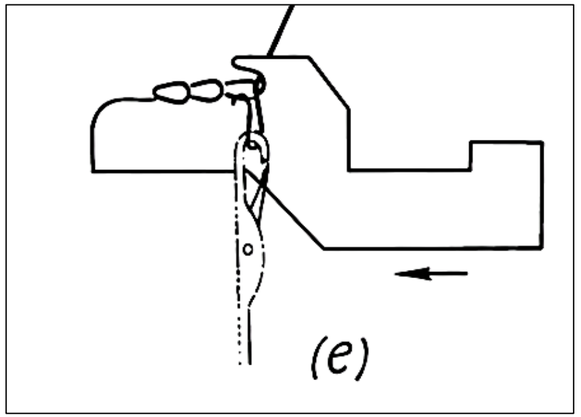
- Joining is the step where the *new yarn meets the old loop.*
- The *new yarn comes into contact with the old loop.*
- This happens *at the beginning of the casting-off process.*
- It helps *connect the new loop with the previous stitch.*

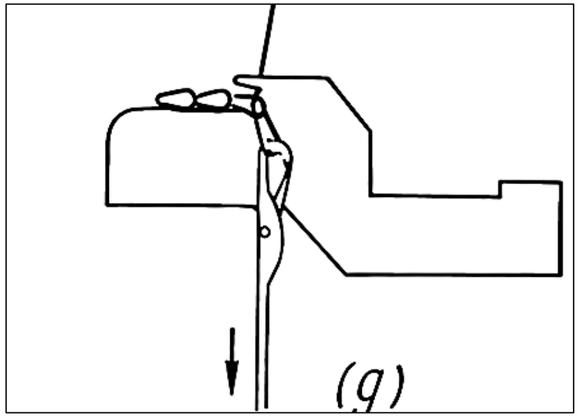


KNITTING CYCLE OF LATCH NEEDLE

Loop Forming (e & g)

- Loop forming is the *step where a new loop is made.*
- The needle *pulls the new yarn through the old loop.*
- The *sinker chin holds the old loop in its position.*
- The needle moves downward and creates a new loop.
- *If the needle goes lower, the new loop becomes longer.*
- *A longer loop means lower stitch density*

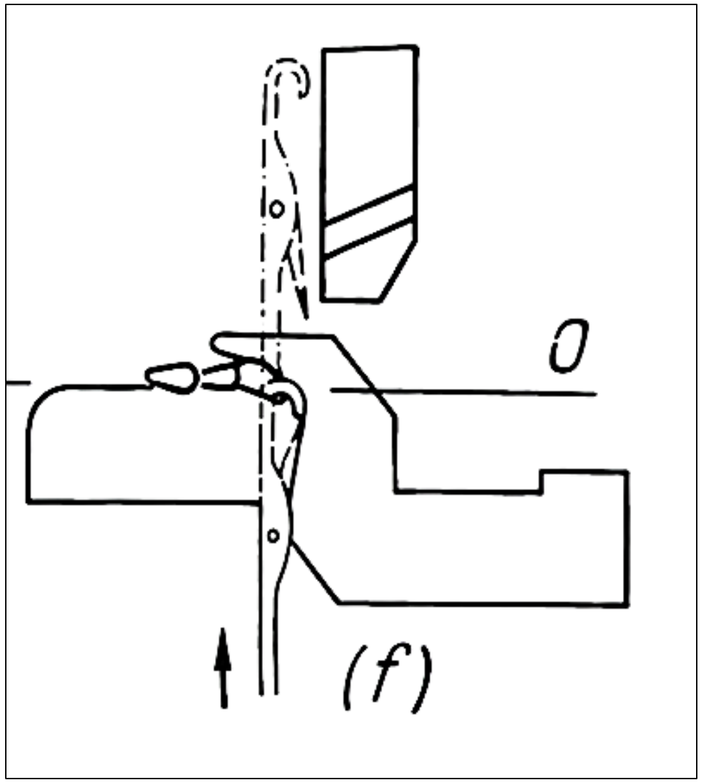




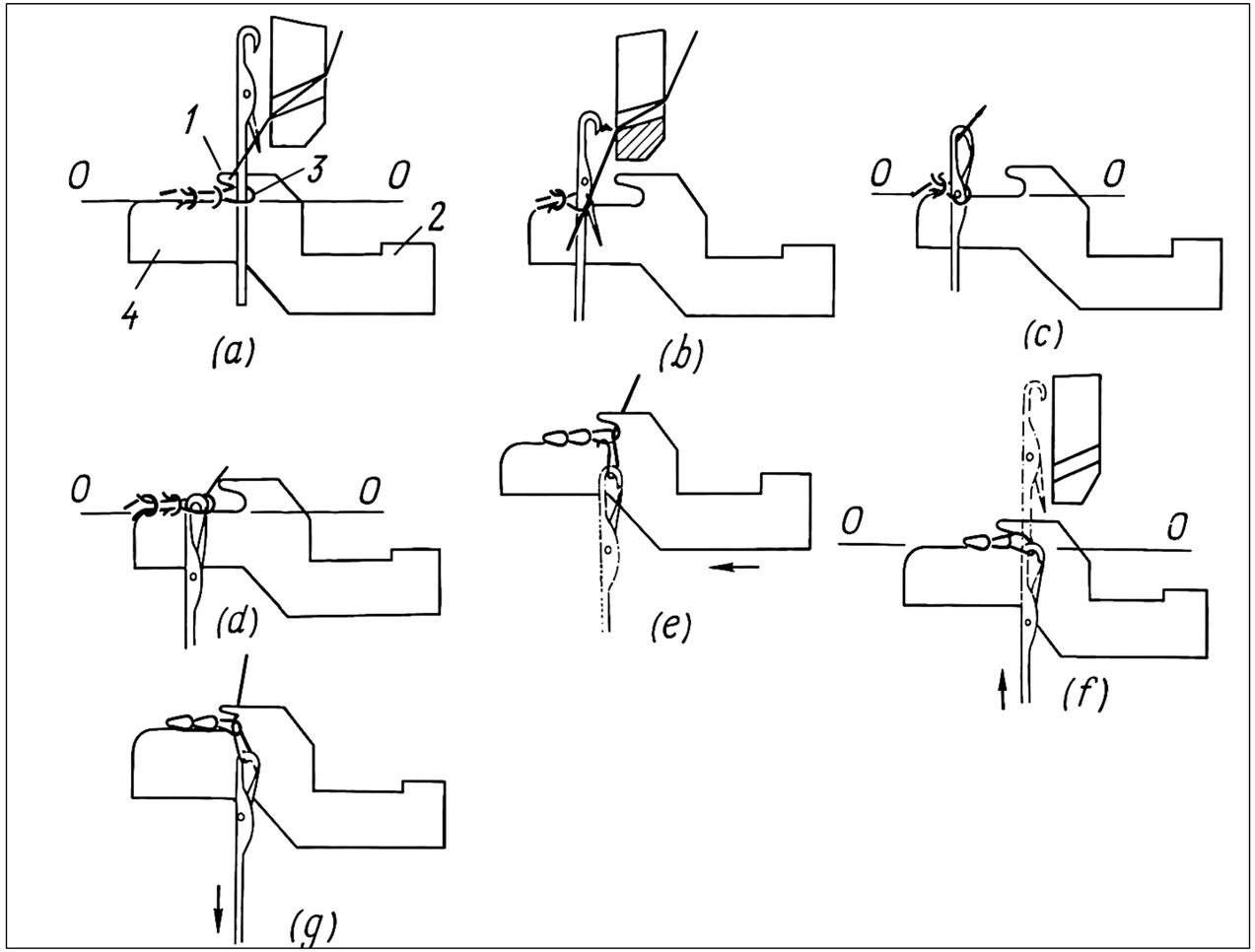
KNITTING CYCLE OF LATCH NEEDLE

Loop Draw Off (f)

- Loop draw-off is the step where *loops are pulled away from the knitting area.*
- The *old loop is moved behind the needle.*
- The *sinker throat pulls the new loops downward.*
- The *sinker chin helps remove the old loops from the knitting zone.*
- The machine's *take-down mechanism helps pull the knitted fabric down.*



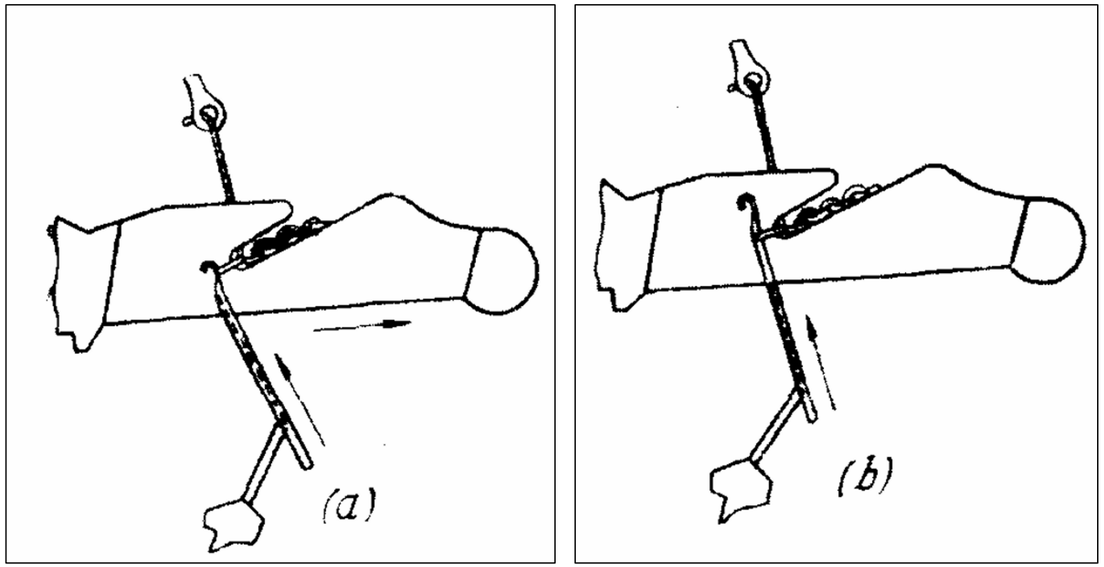
KNITTING CYCLE OF LATCH NEEDLE



KNITTING CYCLE OF COMPOUND NEEDLE

Clearing (b)

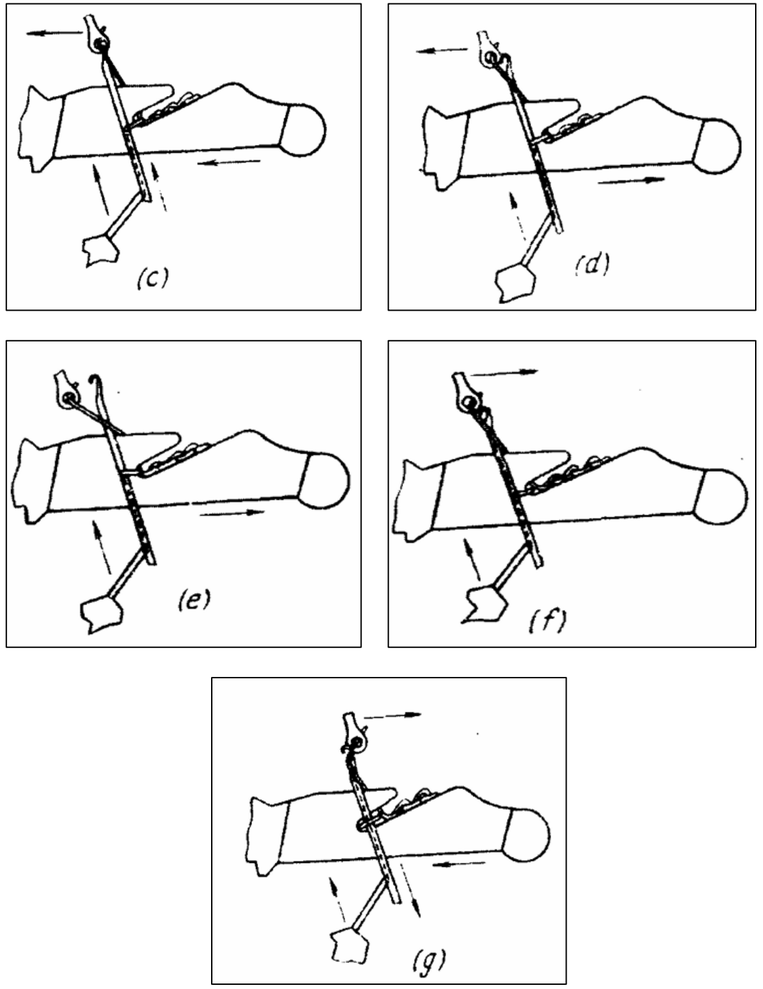
- Clearing is the *first step of stitch formation.*
- The needle *moves upward from the bottom position.*
- The *new loop stays under the needle hook.*
- The *guide bar moves sideways (racks) to the correct position for making the next loop.*
- *The sinker holds the knitted fabric so it does not move upward with the needle.*
- The needle reaches the correct position for receiving new yarn.



KNITTING CYCLE OF COMPOUND NEEDLE

Overlapping or Yarn Laying (c - g)

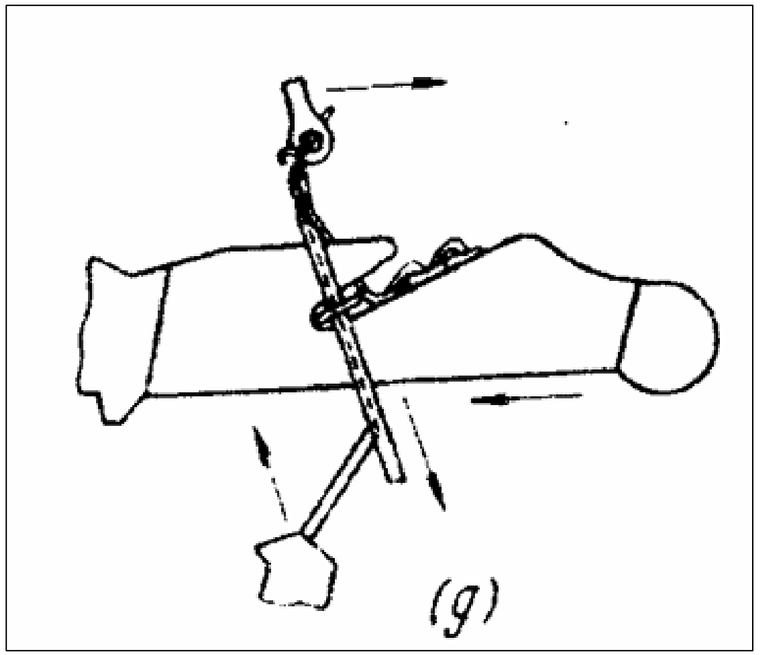
- Overlapping is the *process of placing new yarn around the needle hook.*
- The *needle rises* and the *guide bar starts moving across the needles.*
- The *guide needle passes between the needles* and *places yarn around the needle hook.*
- The *guide bar moves sideways to wrap the yarn around the needle* (usually one needle space).
- The *needle then starts moving downward.*
- *Yarn laying finishes when the needle starts descending.*



KNITTING CYCLE OF COMPOUND NEEDLE

Underlapping (g)

- Underlapping *moves the new yarn under the needle hook.*
- The *needle continues moving downward.*
- The *guide needle moves across the needles.*
- The *new yarn is held by the sinker nib and does not move with the needle.*
- The movement of the guide needle *brings the yarn under the needle hook.*
- The *sinker moves back toward the machine center.*



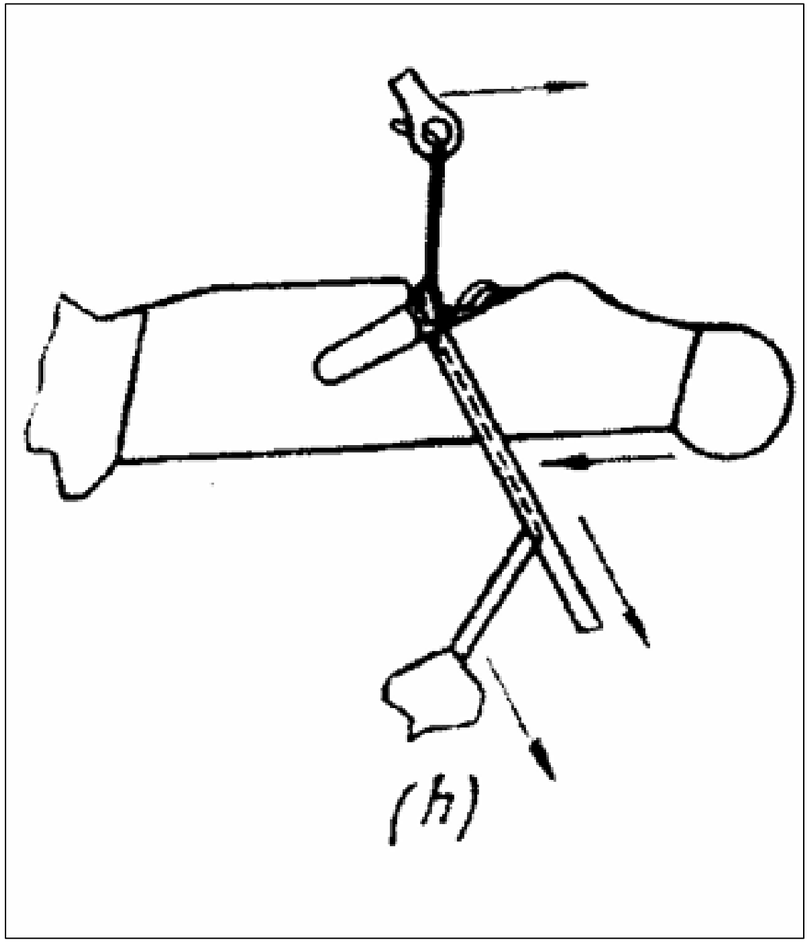
KNITTING CYCLE OF COMPOUND NEEDLE

Pressing (h)

- *Pressing closes the needle hook.*
- *The needle moves downward.*
- *The sliding latch moves over the needle hook and closes it.*
- *The sinker pushes the old loop toward the closed hook.*

Landing-over (h)

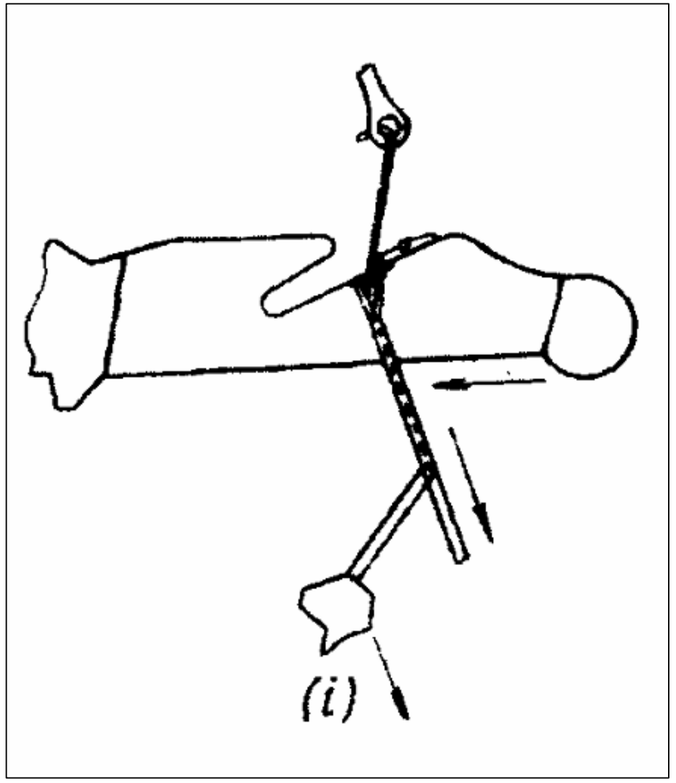
- Landing-over places *the old loop on the latch.*
- The *sinker continues moving.*
- The *sinker pushes the old loop onto the sliding latch.*
- The guide needle moves away.
- The *needle and latch move downward.*



KNITTING CYCLE OF COMPOUND NEEDLE

Joining or Meshing (i)

- Joining connects the old loop with the new yarn.
- The needle moves downward with the latch.
- The *old loop comes in contact with the new yarn under the needle hook.*
- The *guide needle continues moving away.*



Sinking (i)

- Sinking *forms the new loop shape.*
- The *needle moves further downward.*
- The *new yarn under the hook bends and forms a loop.*

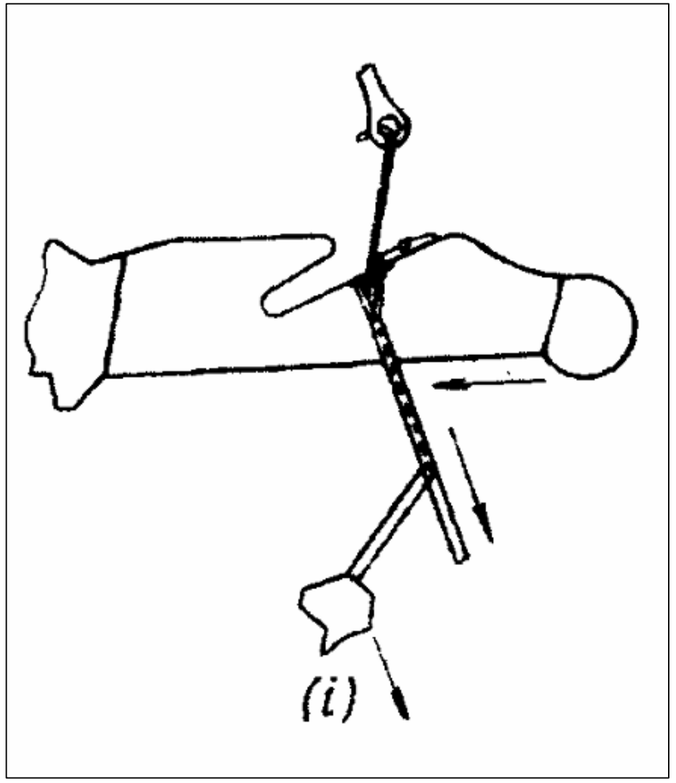
KNITTING CYCLE OF COMPOUND NEEDLE

Casting-off (i)

- Casting-off removes the old loop from the needle.
- The *needle and latch move below the sinker belly*.
- The old loop passes over the new loop.
- The *old loop leaves the needle.*

Loop Forming (i)

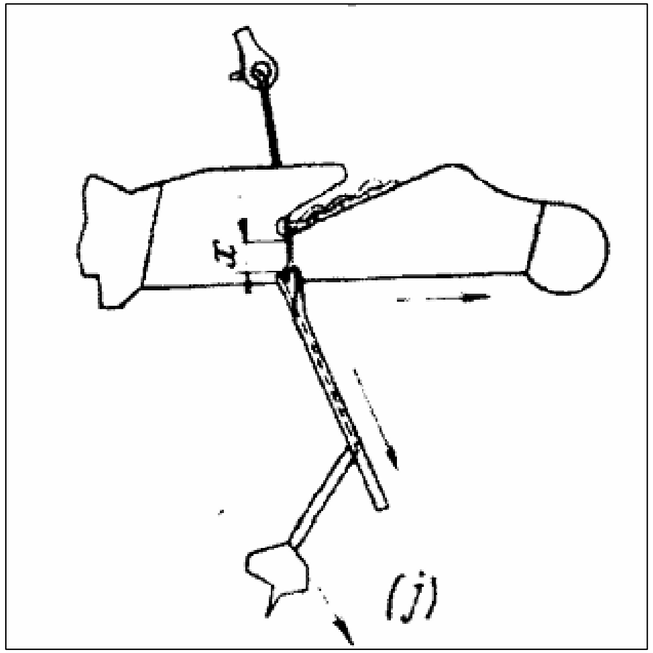
- Loop forming creates the new stitch.
- The needle continues moving downward.
- The new yarn becomes a complete loop.



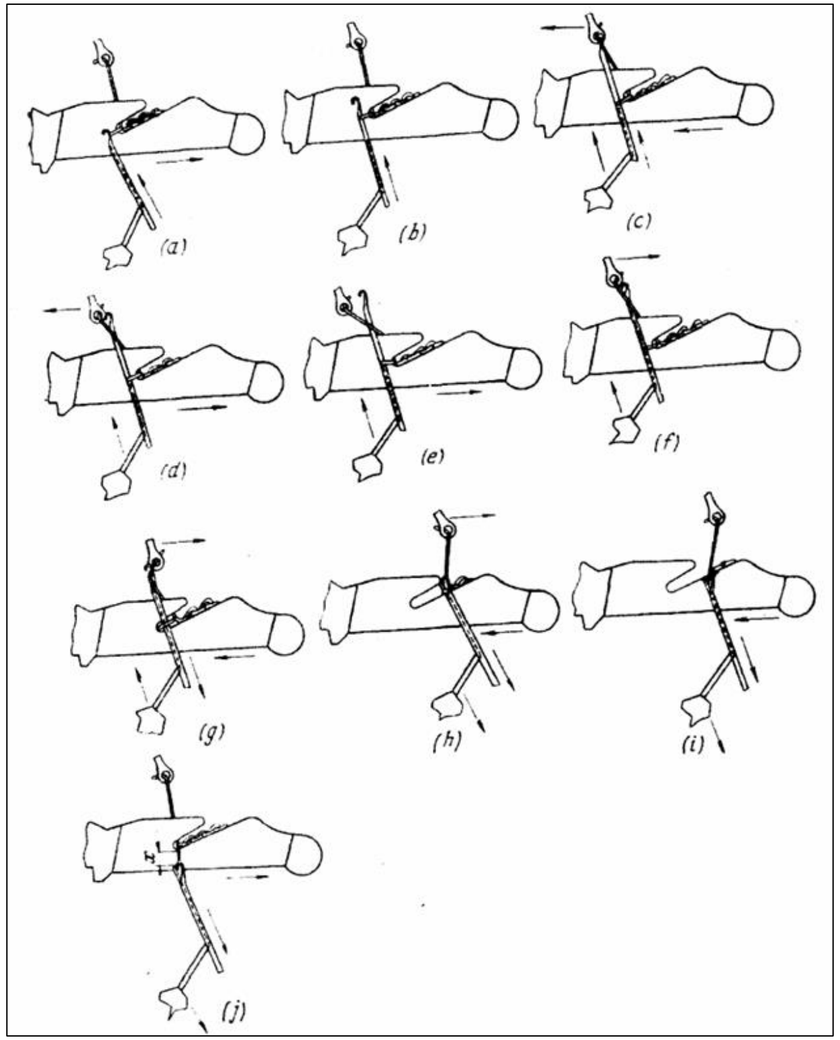
KNITTING CYCLE OF COMPOUND NEEDLE

Draw-off (j)

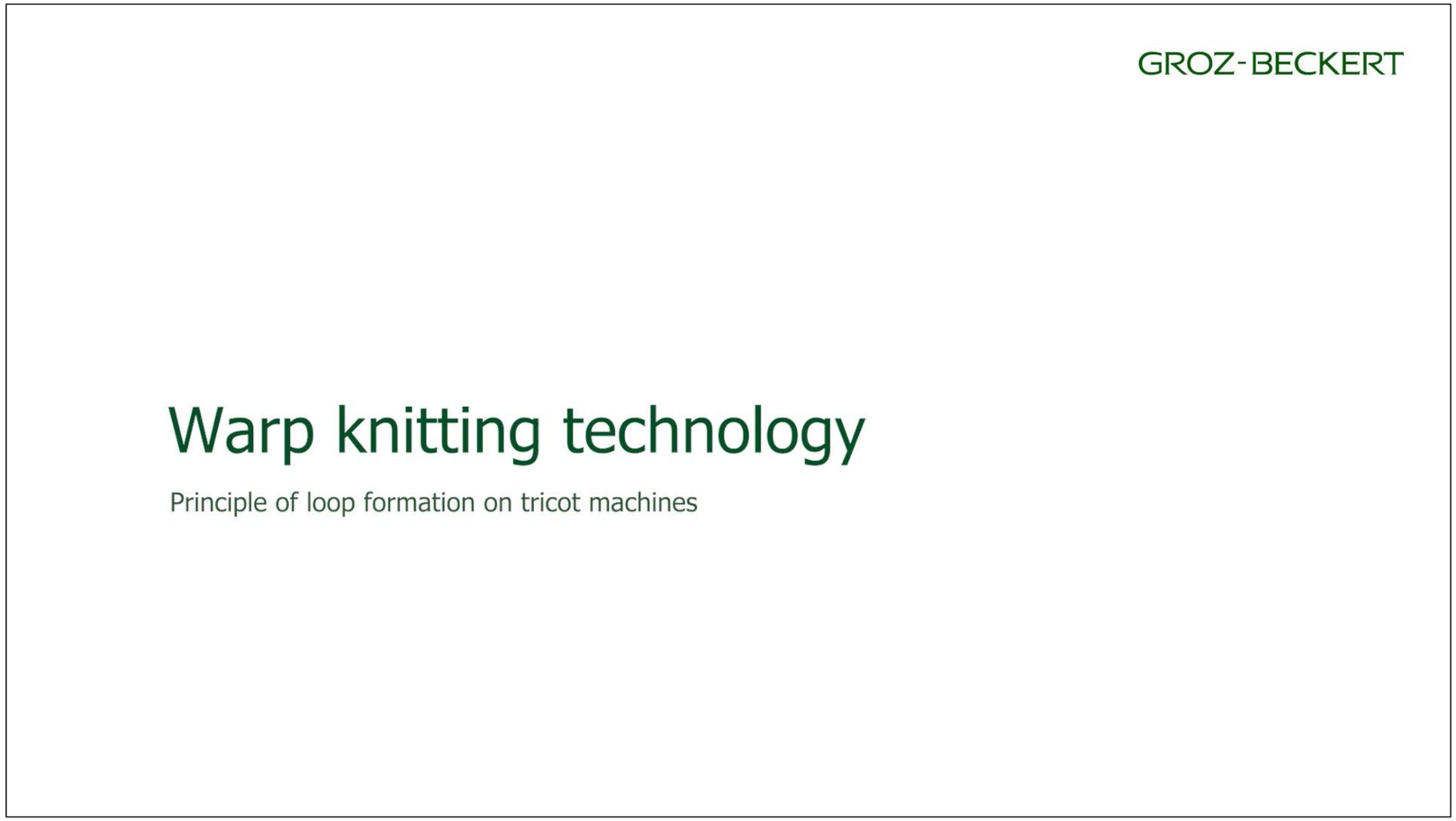
- Draw-off pulls the finished loop away.
- The machine continues rotating.
- The needle starts moving upward.
- The yarn is pulled from the newly formed loop to the guide.
- The knitted fabric moves forward.



KNITTING CYCLE OF COMPOUND NEEDLE



KNITTING CYCLE OF COMPOUND NEEDLE



THANK YOU !

